CATEGORY

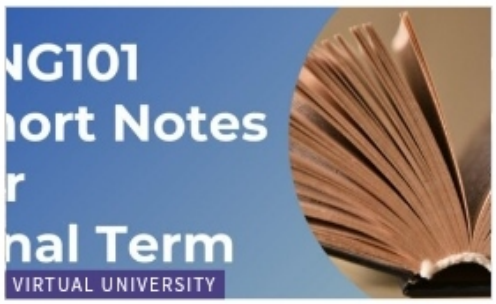
Virtual University

Pak301 Files for Final Term | 100% Prefect Files

Usmanmani - March 8, 2022

PHY101 Notes for Final Term Papers By Concepts Builder

UzmaAameer - March 6, 2022



Eng101 Final Term Notes By Concepts Builder

Ayesha Mustafa - March 5, 2022

ISL202 Complete Handouts For Vu Students

UzmaAameer - February 21, 2022

Latest news

Role of Technology in Education in 2022 | Technology Importance

Pak301 Files for Final Term | 100% Prefect Files

PHY101 Notes for Final Term Papers By Concepts Builder

How Internet Play Key Role in Students Life?

Eng101 Final Term Notes By Concepts Builder

Levels of Responses:

- Simplest response is a reflex action. which is already program, whenever we want to do this thing and that. This is kind of reflex action or preplaned thing which has been built in the Robot.
- Intelligent Action - Adjust actions according to environment knowledge so, Agent can be made in a way that is see in this situation what I do

Example:

The Process of throwing a baseball is largely a reflex action but determining how and where to throw the ball - require knowledge of the current environment In some cases an agent responses improve over time as the agent learn from the environment. This could be taken from the form of developing procedral knowledge. (learning how) or storing declarative knowledge (learning what).

Module # 172

Research Methodologies

AI Domain

AI domain can be persued along two path.

- Engineering Track
- Theoratical Track

Engineering Track :

- Researches are trying to develop systems that exhibit intelligent behaviour and that behaviour already existed in the theoratical research and they are trying to implement behaviour in a perfect way.
- It performed a oriented-methodology.
- Engineer produce a product that meets certain performance goal.

Theoretical Track

- Researches are trying to develop a computational understanding of human intelligence.
- It leads to a simulation-oriented-methodology. so they are inventing something, they are advancing in the knowledge of artificial intelligence and then they are checking whether it is perfect or not in a simulation environment because the underlying goal is to expand our understanding of intelligence and thus the emphasis is on the underlying process rather than the exterior performance.

Example Linguistic Vs NLP

Linguistics are interested in learning how humans process language and thus tend toward more theoretical pursuits.

Natural Language Processing are interested in developing machines that can manipulate natural language and therefore lean in the engineering direction.

Example - Building Shell For an OS.

If we want to build shell for OS. Goal recieve instructions from the outside world through verbal English commands.

Shell need not to be aware of English language as per the engineering or programming they are expecting. They are not interested that this particular agent should be able to know all meanings of copy that's a verb or a noun. So, this is satisfatory for engineer. It can distinguish command of copy, paste, find, search.

However, this is not a aesthetically pleasing to a theoretician. So a theoretical person are not very pleased from this. They want that this agent should be to distinguish this and this even outside of the domain of the that particular problem.

Module # 173

The turing Test

If you built an AI model and you have implemented an AI model in some language and we want to see whether you have come up with an intelligent agent, that is corrved, accurate or not. so this is the test which can tell us thats whether it perfect, accurate or inaccurate.

History:

In the past, Turing test proposed by (Alan Turing) in 1950 has served as a benchmark, as a standered. Everyone was comparing, and utilize this stander to test whether AI model or the system is perfect, accurate or not. However, it is used in the field of artificial intelligence.

How it Work:

Turing's proposal was to allow a human, whom we call the interrogater, to communicate with the test subject by means of a typewriter so human is sitting and his writing a commands and on the other hand they could be a human or a intelligent agent.

- So if the actual human who is typing a command remains unable to distinguish when there was a human and when there was a machine or a intelligence agent. This means that you have made a perfect and a accurate AI system.

Turing Prediction

Turing predicted that by the year 2000 machines would have a 30 percent chance of passing a five-minute turing test. This means you are communicating constantly for five minutes to a machine and you will not realize this for machine and was a human He proposed in 1950, and after 50 years 30 percent of system would be doing perfect. and would be able to pass a test.

- That turned out to be surprisingly accurate.

Today Turing Test is Not used:

Turing test not used today due to following reasons

- This is due to the fact such intelligence has been found to be added with much easier. e.g

Docter Program

You say this program that I am tired

today and it possess another question to you that "why you are tired today" So, this kind of communication goes on until the agent is unable to realize that what you are said so he is not understanding. In that situation, agent say you go on, carry on

- Most recent examples of turning test "successes" include internet viruses that carry on "intelligent" dialogs with a human victim in order to trick the human into dropping his/her malware guard.
- Humans competing against the computer e.g chess, often experience the sensation that the machine possesses creativity and even a personality. So there are many people who are playing chess with a computer and algorce you know that the computer or AI have been built by the humans into the game and sometimes it feeled, this is not an intellegent agent in fact this is a human and this has really a personality and you feel very grate about it.

Module # 174

Understanding Image

Image Processing Simplest Approach

- One of the simplest approach that could be assume that what appear always is an image containing the digit 1 to 8 in a well-organized pattern.
- Picture of Puzzle has been encoded in terms of bits in the computer's memory, with each bit representing the brightness level of the particular pixel.
- Our machine can detect which tile is in which position by comparing the different sections of the picture to prerecorded templates.

Matching Geometric Characteristics.

A technique of recognizing images is one method used in optical character recognition.

Problem: Drawback require a certain degree of uniformity of the style, size and orientation of the symbols being read. You need to write your puzzle in one way. If you change your puzzle game so then you are not able to understand a that pattern and similarly if this technique is being used for a

174

handwritten material when people write 1 in one direction, in second direction, in different ways, in different styles, so in that situation this simplest method is not going to be useful. In such a situation we need to match geometric characteristics.

How it works ?

It work in two steps

1. The first is to extract the features from the image being processed.
2. The second is to compare the feature to those of known symbol.

Process of Understanding Images

Image Processing

It refer to identifying characteristics of the image known symbols.

Image Analysis

It refer to the process of understanding what these characteristics means

REDMI NOTE 8
AI QUAD CAMERA

Module # 175

Language Processing

1. What is the perception of scientist who started work on language processing?

They were able to make the compiler as which were able to translate the programming languages into machine languages. They were considered that it is very near that we could also translate the human language or natural language in similar way as we have translated the high level programing language.

However, this was not reality.

Reality

- Although formal language is differ from natural languages like English, urdu, german only a few years away
- But a programming language are constructed from well-designed primitives so that each statement has only one meaning.
- But in a natural language have a multiple meanings depending on it's context or even the manner in which it is communicated.
- To understand natural language, humans rely on heavily additional knowledge and context.

REDMI NOTE 8 AI QUAD CAMERA

Understanding NLP

To understanding a NLP we require a
Syntactic Analysis
Semantic Analysis
Contextual Analysis

Syntatic :

Major component is parsing.

"Mary gave John a birthday card."
Subject = Mary

"John got a birthday card from Mary"
Subject = John

Semantic :

In semantic analysis we want to identifies the meaning Here, we need to identify the role of each word in statement.

"Mary gave John card" and "John got a card from Mary."

These both are same thing if we able to perform a semantic analysis in accurate and a correct way.

Contextual :

Context of the sentence is brought into the understanding process.

"The bat fell to the ground" has different meaning in following two context :

baseball game
Cave exploration

REDMI NOTE 8
AI QUAD CAMERA

Module # 176

CS Impact On Society

CS has both positive and negative impact on society

Positive impact

- It has improved the communication. You are able to communicate sitting in Pakistan to a friend Europe or America. It is a CS that has made the network and internet through which we can communicate.
- It has increased access to educational information via the internet. e.g In Virtual university, you are studing using internet by sitting in your home. So you do not need to move a place of university where the situated actually. It means Education is on our finger tips that we can view, search, identify about our particular thing/needs.
- It increased the productivity. So initially we were only able to write mainly a few pages but today we can write similar thing online and online it can be done very rapidly.
- It has exponential business growth. You can well managed the buisiness you can administer, what are the

basically products being sold, how your employes are working.

- Faster response times e.g emails, instant messaging and chat. Today we can send a messages in a different ways.

Negative Impacts.

- It has ties to increased obesity - because you are sitting, you are not working very hard. you are not working mainually Always sitting increase the chance of obesity.
- It has increased crime and access to private information. So all type of online transaction and online theft has started.
- Today, has bank accounts are available online, you can hack data account and can transfer all money into your own account.
- Potential over-dependence on a technology. So it has already increased.

REDMI NOTE 8 AI QUAD CAMERA

Module # 177

CS Impact On Health

CS has both positive and negative effect on our health.

Positive Impact

- Eassy access to online health centre If you want to search near by hospital a very known docter in world or region. So you have access to that particular thing.
- By using a CS we can take online appointment by sitting at home.
- We can also recieve report email at home. So if you have a some undergone disease or some problem and you go to a hospital, they take your blood and then perform a some experiment. That report can be accessed to email by sitting at home. You are not need to go a hospital.
- Hospitals can be managed more efficiently. So the hospital administrations can managed those hospitals in a way that how many people visited patient in a hospital and how many employes are working What are their salaries, what are their earning, what are the

REDMI NOTE 8
AI QUAD CAMERA

profits.

- We can also get online health tips.
- We can also learn the health news by using a web.

Negative Impact

However, this have some negative impact on health.

- Computer wrists, This is a phenomenon when you are typing too much, then there is a pain in your wrist.
- There is a eyesight problem Our eyes are not look 2D images. So our eyes have built by God to see only a 3D images. The radiations coming from screen can also affect the health.
- If you are hearing something in your ears you get a hearing loss. as well because the voice going inside drum in our ear creating a disturbance and have a different kind of loss in ear.
- We are socially isolated. Whatever we want to do at home we are not meeting with anyone.
- Communication issues because in a isolation you are not communicated with people, when you are going outside for some real work. you have experienced some communication Problem

- Posture problem when you are sitting in a wrong position while using a computer. that could lead you a bone issues. in your body.
- Radiation demage your mind. If you are using a smart phone or even any phone that if you remaining in that phone all the night and it near your bed it emit the radiation. So it is better that you put a mobile in aeroplane mode.

Module # 178

CS Impact On Environment

CS has both positive and negative effect on environment.

Positive Impact

- Paperless - Online reading and publishing. So we are not using most of the time paper and this means we are saving trees / tons.
- American use about 680 pounds of paper per person, per year.
- This amount to about 85 million tons of paper or 2 billion trees.
- The average american household throws away 13,000 pieces of Paper (around 1 billion trees in total in USA) each year. Most in the form of Packaging and Junk mail.

When we using cs so it have a direct effect on environment. to save the environment and trees to making environment clean.

Negative Impacts

- In offices, people print unnecesarly material. Many people have habit, that they are not able to read things online. So they prefer that would print each and every thing and then read. In such a situation, it is complicated.
- Computers are made from a heavy metals (Lead, Mercury, Beryllium, Cadmium, PVC) when thrown away these all things cause water - contaimination and air pollution
- E-waste is sent to developing countries where people extracts material from these electronics such as gold, silver and copper.
- They extract these things by burning the substances and that processes releases hazardous smoke into the air.
- The people Present around the e-waste gain diseases such as skin cancer, brain demage, lung cancer and kidney disease.
- It also use lot of energy.

Module # 179

Ethical Issues

Ethical:

It is relating to a moral principles or the branch of knowledge dealing with these

Major Issues

Privacy
Digital ownership
Data gathering

Privacy:

Capacity to store, organized and exchange records.

For example:

You have personal data and you putting your data in emails or passward. Those data are passing through different roots. So whether these roots saving your emails or your passward, your data or not.

Digital Ownership

Digital information is flowing more freely We know a digital information in different capacity are moving.

Legal notions such as copyright have struggled to keep up with the digital era.

Data Gathering:

The united state has passed legislation allowing the government to actively monitor private citizens in the name of a national security. This means they are saying we are the national security agencies and we have the complete priviliges to access a data.

Module # 180

Software Licenses And Information Privacy

Software Lisence:

A software license is a document that provide legally binding guidelines for the use and distribution of software. So you might have use or you might install some of software on your computer and that software might have come with the software license. Actually, in that document there is a written statements that tell you whether you can use this software legally or whether you can distribute that software to some other people in your organization.

REDMI NOTE 8
AI QUAD CAMERA

Software license typically provide and users with the right to one or more copies of the software without o violating copyrights.

- Terms And Conditions
- Fair use of the software
- The limitations of liability
- Warranties and disclaimers which are mentioned at the different bases of the software.
- There are mention as well the protection If the software or use violates the intellectual property rights of other.

Software License Types

Proprietary

Free

Open source

Proprietary

In proprietary software license you are not allowed to use that software out of the boundary

Free

In free software license, you are allowed to use the software the way you want.

Open Source

In open source software license basically A software that has been built by some one in some powers or the initials ways and then that

person shared software globally with all of the user in the world.

Information Privacy:

It also called a data privacy. This deal with ability an organization or individual has to determine what data in a computer system can be shared with third parties.

Some Acts

- COPPA
- HIPPA
- ECPA
- Vedio privacy protection act
- Gramm-Leach-Bliley act

Modulle # 181

Intellectual Property (IP) is an ideas, information and knowledge and this is called intellectual because it is creative output and property because it is viewed as a tradeable product If you have any idea and it is a unique and you don't want to share this idea with others so you can use the IP in different forms to protect your idea.

REDMI NOTE 8
AI QUAD CAMERA

Intellectual Property Rights (IPR) are specific legal rights which protect the owner of IP.
There are different ways to protect our ideas. we can protect our ideas by using patent.

Patent:

A patent is a legal monopoly lasting 20 years granted in exchange for describing an invention and paying fees to the patent office. So, normally patent office is in a USA. If we use patent so for next 20 year no one can use our idea.

- Patent is destroyed when we publish the application.

Copyright

Next protection is copyright. Copyright is also a usual way of protecting software. although some software may be patented if it is a functional part of an invention.

Database Right

Database right are apply to databases which are not protected by copyright.

Design Right

Design right applies to aspects of the shapes of an article. Unregistered design right (which cover

computer chip so those type of thing can be patent they cannot become copyright, they can go with the design right

Trade mark:

A trade mark is a mark or other distinctive sign applied to or assosiated with products or services

Confidential Information

Confidential Information is knowledge which only you possess and which you have only revealed under a non-disclosure agreement

Module # 182

Computer security also known as cyber security OR IT security. It is the protection of information systems from theft or demage to the hardware, the software and to the information on them

Security areas:

Confidentiality:

It means that only authorized users can access the data resources and information

Integrity:

Only authorized users should be able to modify the data when needed.

REDMI NOTE 8
AI QUAD CAMERA

ConceptsBuilder.com

Availability

Data should be available to users when needed.

Authentication:

Are you really communicating with whom, you think, you are communicating with?

Why its Important:

- It prevent a data theft such as bank account numbers, credit card information, passward, work related documents, sheets etc.
- Data present in a computer can also be misused by unalthorized instrusions.
- Malicious intents can also be factor in computer security. So Introders often use your computers for attacking others computers or websites or network for creating haroc.

Firewall:

This is basically something which protect you from some intext security. A firewall is a security consious piece of hardware or software that sits between the internet and your network with a single-minded task. It means no one is able to come into your area.

REDMI NOTE 8
AI QUAD CAMERA